PID grid sweep

Wall-E balance controller • MuJoCo 3.7.0 • 13 October 2026

Bottom line

No tested gain combination balanced the current simulated model. All 52,521 candidates fell in the initial trials. The top-ranked candidate failed again in longer validation, so there is no winning PID setting to recommend.

Search definition

Parameter	Values tested	Count
Kp	0 to 30, step 0.5	61
Ki	0 to 1, step 0.05	21
Kd	0 to 20, step 0.5	41
Total	61 × 21 × 41	52,521

Setpoint fixed at 0° (the configured reset-pose zero); firmware-clamped PWM; 2 s maximum initial trial; 30 top-ranked candidates rerun for up to 10 s. A fall was total chassis tilt reaching 45°. This exhausts the stated grid only—not continuous gain values or all possible setpoints.

Best-ranked point (not a recommendation)

Kp = 19.5, Ki = 0.75, Kd = 20.0. Survival was 0.20 s in both the initial run and 10 s validation. Validation stopped after falling: peak absolute pitch 48.91°, pitch span 48.65°, true-pitch RMS error 23.59°, and mean absolute PWM 71%.

What this means

This diagnoses the current simulated plant/controller setup, not the real robot: gains cannot compensate for incorrect actuator polarity, insufficient or unrealistic torque, bad contact, or incorrect dynamics. The sweep used an artificial constant 0.02 N·m full-scale torque and assumed wheel signs left +1/right −1. Neither is a measured motor model or verified hardware direction. The 60% minimum PWM clamp can also make small nonzero commands jump to substantial effort.

Do not flash these top-ranked values as a hardware solution. First inspect the saved telemetry and verify the restoring command direction, wheel/floor contact, and measured motor response; then rerun tuning against a validated plant.

Artifacts and reproducibility

Full data: experiments/mujoco_pid_grid/20261013T_full_firmware_grid/ (grid_results.csv, validation_results.csv, best_candidate_telemetry.csv, run_metadata.json). Editable report source: docs/MUJOCO_PID_GRID_REPORT.md.

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.venv/bin/python -m simulation.diagnostics.sweep_balance_pid_grid --kp-range 0 30 0.5 --ki-range 0 1 0.05 --kd-range 0 20 0.5 --trial-seconds 2 --validation-seconds 10 --validation-top 30 --setpoint-deg 0 --pwm-output-mode firmware_clamped --ideal-test-torque-nm 0.02 --left-sign 1 --right-sign -1 --output-dir experiments/mujoco_pid_grid/20261013T_full_firmware_grid
```

Verification after the sweep: 14 tests and 6 subtests passed; simulation.verify_balance_io passed. These checks validate software plumbing only, not physical balancing.